The explicit coefficient series and the phase-derivative dictionary (Taylor-tree Stage 5)

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Abstract

Units 250–254 close review v21's qualification (ii): the family coefficients $A_{\mu,j}(\xi,\eta)$ of Stage 4, defined as limits of polynomial truncation coefficients, are identified with the paper's explicit absolutely convergent series. With J the constant-free part of the coefficient family of ξ and $c * e$ the Cauchy product $(c*e)_\gamma = \sum_{\alpha\le\gamma} c_\alpha e_{\gamma-\alpha}$,

$$A_{\mu,j}(\xi,\eta) = \sum_{p\ge 0} \frac{\beta^p}{p!}\, T_{\mu,j,p}\big(c^\eta * J^{*p}\big),$$

$$T_{\mu,j,p}(f) = \prod_i \frac{1}{2k_i} \sum_\gamma f_\gamma \sum_{q=j}^{d-1} c_{\mu,q}(u^{h+\gamma}) \binom{q}{j} \int_0^\infty t^{\mu-1}(-\log t)^{q-j}(\sqrt{t})^p e^{-\beta t+\beta\sqrt{t}a}\, dt$$

(Headline XXIX, `familySpectralCoeff_eq_series`, $\mu > 0$), where $(c^\eta * J^{*p})_\gamma = \sum_{\mathbf{m}+\mathbf{n}=\gamma} \frac{\eta_{\mathbf{m}}}{\mathbf{m}!}\xi_{\mathbf{n},p}$ is the paper's collected coefficient (eq:flucttreeterms) once the families are normalised Taylor coefficients. The phase half of the derivative dictionary is a theorem: $\partial_a^p S_\mu(a) = \beta^p \int_0^\infty t^{\mu-1}(\sqrt{t})^p e^{-\beta t+\beta\sqrt{t}a}\, dt$ (unit 250). Review v22: qualified pass, no mathematical defect. No `sorry` and no additional `axiom` declarations; 9130 jobs.

1 Architecture

Cauchy product without an antidiagonal instance (u251). Mathlib has no `HasAntidiagonal` on $\mathrm{Fin}\, d \to \mathbb{N}$; the convolution is a finite sum over $\mathrm{Iic}\,\gamma$ and the double series over pairs is regrouped along $\gamma = \alpha+\delta$ by the equivalence $(\alpha,\delta) \mapsto \langle \alpha+\delta, \alpha\rangle$ and `HasSum.sigma`, giving $\mathrm{eval}(c*e) = \mathrm{eval}\, c \cdot \mathrm{eval}\, e$ on the cube and $\mathrm{mass}(c*e) \le \mathrm{mass}\, c \cdot \mathrm{mass}\, e$.

Collected coefficients (u252). A monomial list collects to a finitely supported family; list product collects to convolution, powers to convolution powers, the fluctuation part to the constant-free family and box truncations to restrictions. Every Stage-3 list sum $\sum_{(\gamma,c)\in P} c\, f(\gamma)$ equals $\sum_\gamma (\mathrm{coeffFn}\, P)_\gamma f(\gamma)$.

Kernel functional (u253). $\mathrm{coeffTerm}_p(P) = T_p(\mathrm{coeffFn}\, P)$ with $|T_p f| \le K_k D\, M_{\mu,d-1,p}(a)\, \mathrm{mass} f$: the coefficient functional is ℓ^1-continuous, uniformly in the monomial by the Stage-3 budget.

Identification (u254). The truncation coefficients are the same series for the truncated families; each term converges as the box grows (^{1} continuity, the family power-difference estimate $\mathrm{mass}(c^{*p} - c'^{*p}) \leq pB^{p-1}\mathrm{mass}(c - c')$, tail masses $\to 0$), and the series passes to the limit by dominated convergence with the Tonelli majorant $\sum_p \frac{(\beta B)^p}{p!} M_{\mu,n,p}(a) = M_{\mu,n}(a + B)$. Uniqueness of limits identifies the Stage-4 coefficient with the explicit series.

Phase derivatives (u250). Differentiation under the integral (`hasDerivAt_integral_of_dominated_loc_of_deriv_le`) with the log majorant at phase parameter $|a| + 1$ gives $\partial_a\,\mathrm{fluctMoment} = \beta\,\mathrm{fluctMoment}$ at the next phase order, hence $\partial_a^p S_\mu = \beta^p S_{\mu+p/2}$ (the paper's displayed formula omits β^p).

2 Status of the programme

Proved: the Taylor-tree expansion and remainder on general boxes for phase and amplitude given by absolutely summable power series (Headlines XXVII–XXVIII), with the paper's exponent set, logarithmic indexing and cutoff-independent coefficients equal to the paper's explicit series (XXIX). Not proved: the Cauchy-estimate bridge from holomorphy on a polydisc to summability of the Taylor coefficients (no several-variable Cauchy estimates in Mathlib; Astra #29 estimates 15–25 units and does not budget it), and the spectral half $(-\partial_\mu)^i$ of the derivative dictionary (next unit).

3 Lessons

Dot notation on hypotheses of a `def`-alias type (`AbsSummable`) resolves to lemmas in that namespace first; call `Summable.add` explicitly. `conv` already existed in the project namespace: qualify as `CoeffFamily.conv`. `Pi.sub_apply` before `Finset.sum_sub_distrib`; `List.filter_cons_of_pos/neg` for filters; `Finset.sum_coe_sort` for fibre sums over a Finset coercion; the pi-type order instances live in `Mathlib.Data.Pi.Interval` and `Mathlib.Algebra.Order.Sub.Prod`.